

like the railway locomotives, bridge superstructures were imported from England to the worksites. Here engineers would supervise assembly. As a result, bridge engineers working in India were called “Meccano Engineers”, after the tiny sets used by British children to make toy bridges.



▲ **The two-line Tannah Viaduct** in western India was made using both masonry arches and an iron bridge section to aid in navigation. This picture was taken sometime between 1855 and 1862.

The processes

The substructures for river bridges were usually fixed using either the screw pile or the well-foundation method. The former was used while bridging smaller water bodies with calmer currents although the BB&CI Railway utilized them on many major rivers between Bombay and Ahmedabad. These bridges needed cast iron piles screwed deep into the clay layer underneath the riverbed. One such bridge was the original Vasai Creek Bridge. Built in 1864, its foundations were screwed 6–22 m below bed level with the help of oxen-rotating poles on a capstan attached to the pile top.

Well-foundations were made by sinking open brickwork caissons, (chambers open from both ends), through the silt to rest on clay or rock beds. Wells were fortified with iron work, filled with sand, and capped after sinking. This method was invented in India because of the difficult subsoil conditions of most of the region's rivers. The Old Naini Bridge over the River Yamuna in Allahabad was built this way. It has one distinct pier that juts out. Known as the Elephant's Foot, the pier is built over an almost 16-m-wide stone arch that rests on a manually assembled floor of stone and ashlar, as a solution to well-sinking problems.